Clay Ramey

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LLM-Based Test Case Generation for 21 CFR 117.130

Overview

This report compares test cases generated by two LLMs for five requirements from 21 CFR 117.130. The regulation covers hazard analysis rules for food safety. The two models used were Mistral 7B and Mistral 7B Q4_K_M. The Q4_K_M version is a quantized model. Quantized models use less memory by storing weights at lower precision. Both models were run locally using Ollama. The comparison looks at three things: coverage, completeness, and correctness.

Selected Requirements

The five requirements used for this task came from the group project:

- REQ-117.130-001B – Identify known hazards
- REQ-117.130-001F – Hazard analysis must be written
- REQ-117.130-002A – Biological hazards
- REQ-117.130-003A1 – Assess severity and probability of hazard occurrence
- REQ-117.130-003A2 – Evaluate environmental pathogens for ready-to-eat foods

Coverage

Coverage checks whether both models produced at least one test case per requirement. Both models produced a test case for all five requirements. Coverage was 100% for both models.

Completeness

Completeness checks whether all required fields are present. The required fields are test_case_id, requirement_id, description, input_data, and expected_output. Both models included all five required fields in every test case.

Both models also included the optional fields steps and notes. The steps were useful in both cases. Each model gave three clear steps per test case. The notes field was more useful in the standard model. It referenced specific FDA regulations and listed real assumptions. The quantized model sometimes put placeholder text like "<none>" in the notes field. That is technically present but not useful.

Correctness

Correctness checks whether the description actually matches what the requirement says. Each test case was checked against keywords tied to its requirement. Both models passed all five correctness checks.

The standard model matched more keywords in some cases. For REQ-117.130-001F it matched both "written" and "document." The quantized model only matched "written." For the other four requirements both models matched the same keywords. Both models produced descriptions that tested what the requirement actually says.

Findings

Metric	Mistral 7B	Mistral Q4_K_M
Coverage	100%	100%
Completeness	100%	100%
Correctness	100%	100%

Both models scored the same on all three metrics. The difference shows up in the quality of the content. The standard model gave more specific input data. For REQ-117.130-002A it listed sample types like raw materials and environment swabs. The quantized model just said "sample food product." The standard model also wrote notes that referenced real FDA guidelines. The quantized model's notes were often vague or empty.

Conclusion

Both models can generate valid test cases for regulatory requirements. The scores were equal across all three metrics. The standard Mistral model produced better output overall. Its descriptions were more specific and its notes were more useful. The quantized model passed all checks but would need more editing before being used in a real testing process.